

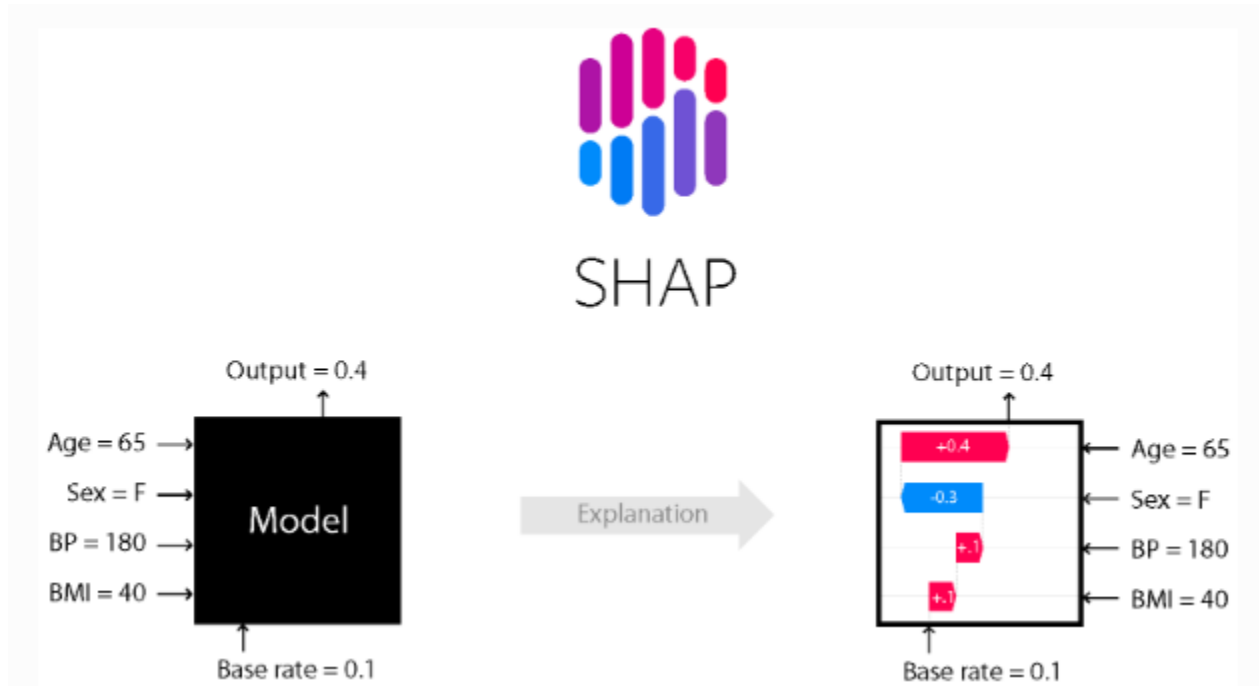


Explicabilidad

SHAP

<https://shap.readthedocs.io/en/latest/>

SHAP

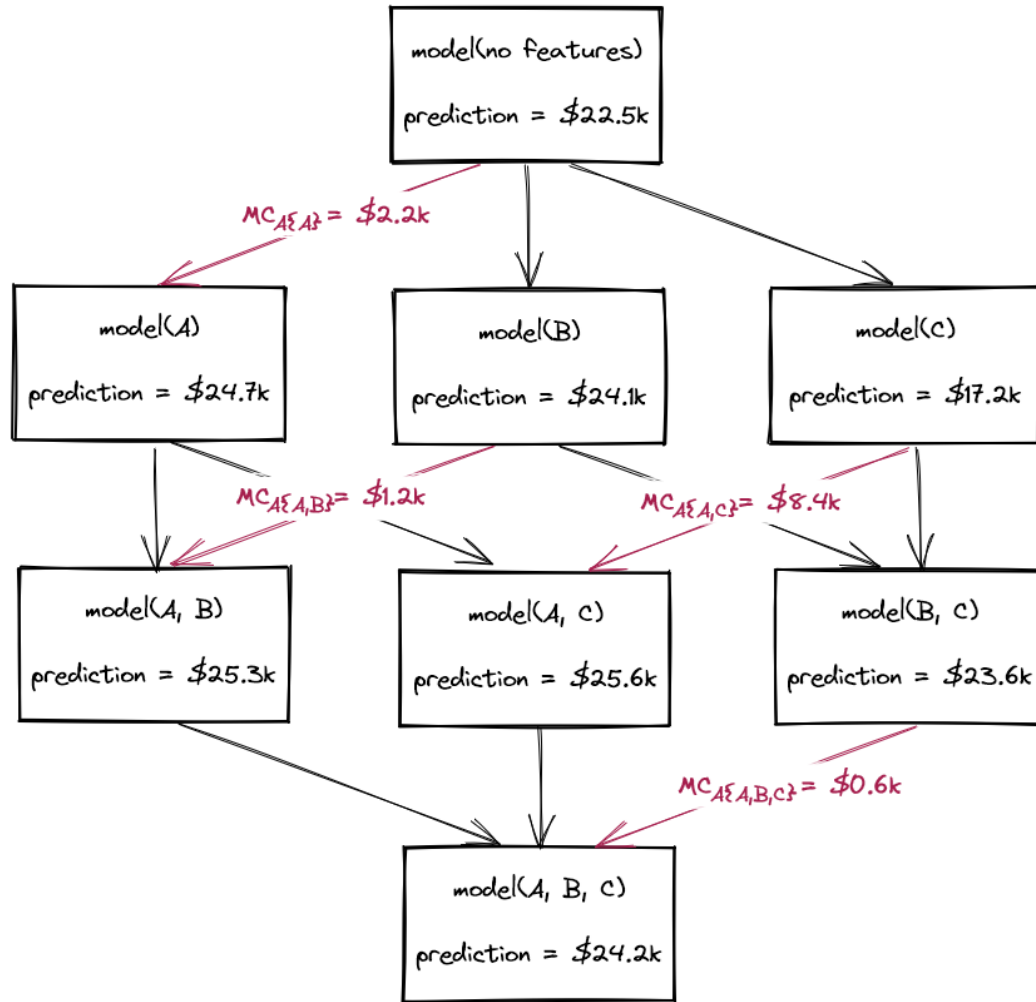




Ejemplo: precios de casas

	A	B	C	predicted price
house1	10	6.5	0.5	\$24,200

$$p(\text{house1}) = b + \text{shapley}_A(\text{house1}) + \text{shapley}_B(\text{house1}) + \text{shapley}_C(\text{house1})$$




$$shapley_A(house1) = 1/3 \times 2200 + 1/6 \times 1200 + 1/6 \times 8400 + 1/3 \times 600$$

Se hace lo mismo para B y C.

$$f(x) = base\ value + sum(SHAP\ values)$$

$$24,200 = 22,500 + 2,550 + 1,250 - 2,100.$$

A y B empujan con correlación positiva, C con correlación negativa.

Para cada dato del conjunto de test se puede calcular un shapley Value de cada feature.



Para modelos de regresión, el valor base es igual a la media de la variable objetivo (por ejemplo, la media del precio de las casas en el conjunto de datos), mientras que para modelos de clasificación, el valor base es la media del logaritmo de chances de la clase positiva.

SHAP **cuantifica** qué tan importante es cada feature a la hora de hacer predicciones. Esto puede ser una comprobación útil para asegurarse de que el modelo se comporte de manera razonable: ¿el modelo utiliza las características que esperamos basadas en la experiencia en el dominio?

Ocasionalmente, SHAP puede revelar una relación inusualmente fuerte entre una característica y el resultado. Puede ser una herramienta de diagnóstico útil cuando los modelos tienen un rendimiento predictivo sospechosamente alto.

En otras ocasiones, estas relaciones sorprendentes son legítimas y pueden servir como un mecanismo para la generación de hipótesis: es decir, ¿por qué esta característica es tan predictiva y qué podemos hacer para validar esta teoría?

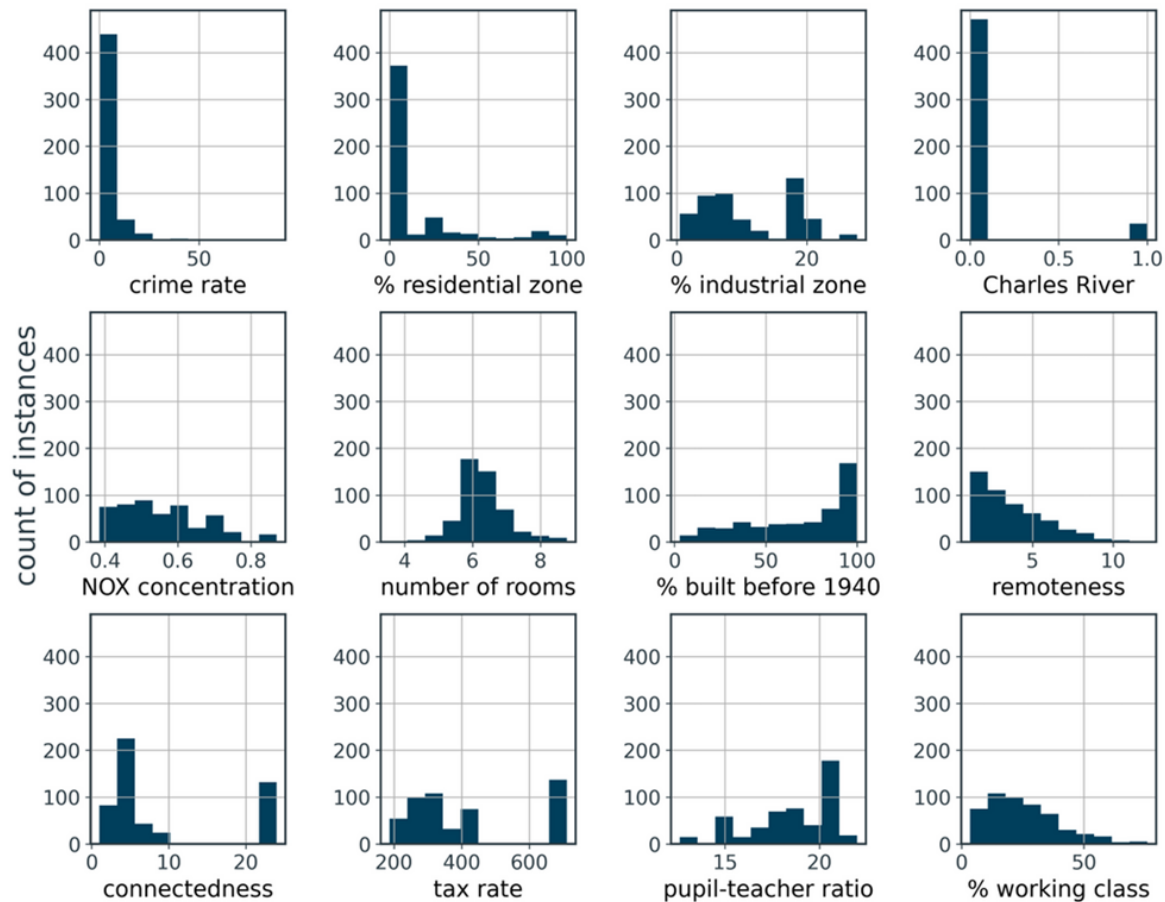
SHAP solo te dice lo que el modelo está haciendo en el contexto **de los datos en los que ha sido entrenado**: no necesariamente revela la verdadera relación entre las variables y los resultados en el mundo real. A menudo, los tomadores de decisiones están tentados a ver las características en los análisis de SHAP como controles que se pueden manipular para lograr resultados específicos, por lo que esta distinción debe comunicarse.

Interpretabilidad local: sobre una determinada predicción

Variable Name	Description
crime rate	Per capita crime rate in the town.
% residential zone	Percentage of land zoned for residential use.
% industrial zone	Percentage of land zoned for industrial use.
Charles River	1 if the house borders the Charles River; 0 otherwise.
NOX concentration	Nitric oxides concentration (parts per 10 million).
number of rooms	The average number of rooms per house in the housing unit.
% built before 1940	The proportion of houses built prior to 1940 in the unit.
remoteness	A measure of how far the housing is from employment centres (higher is more remote).
connectedness	A measure of how good the local road connections are (higher is more connected).
tax rate	Property tax rate per \$10,000 of house value.
pupil-teacher ratio	The pupil:teacher ratio of the town.
% working class	Percentage of the population that is working class.

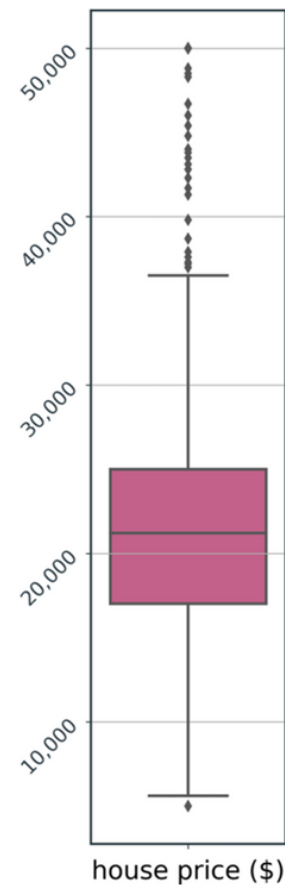
Table 1. The model input variables used to predict house prices. This is a modified version of the Boston Housing Price dataset.⁷ Variable names and descriptions have been simplified.

Model Input Variables



predicts...

Target Variable



The model's predicted house price for this example.

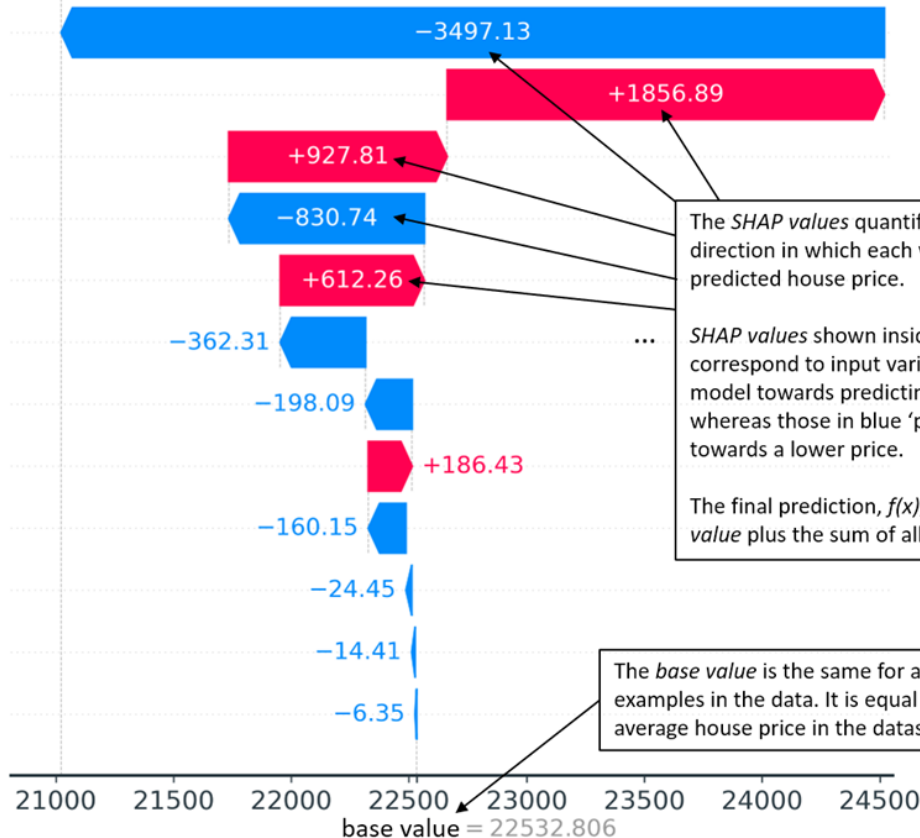
$$f(x) = 21022.57 = \text{base value} + \text{sum}(\text{SHAP values})$$

A **waterfall plot** provides a detailed breakdown of how each input variable contributes towards the predicted house price for a single instance of the data.

These are the input variables, ranked from top to bottom by how much impact they have on the model's prediction for this example from the data.

The grey numbers denote the values of the variables for this particular instance.

5.878 = number of rooms
16.2 = % working class
21.4 = % built before 1940
6.498 = remoteness
0.409 = NOX concentration
345 = tax rate
4 = connectedness
0.058 = crime rate
6.07 = % industrial zone
18.9 = pupil-teacher ratio
12.5 = % residential zone
0 = Charles River

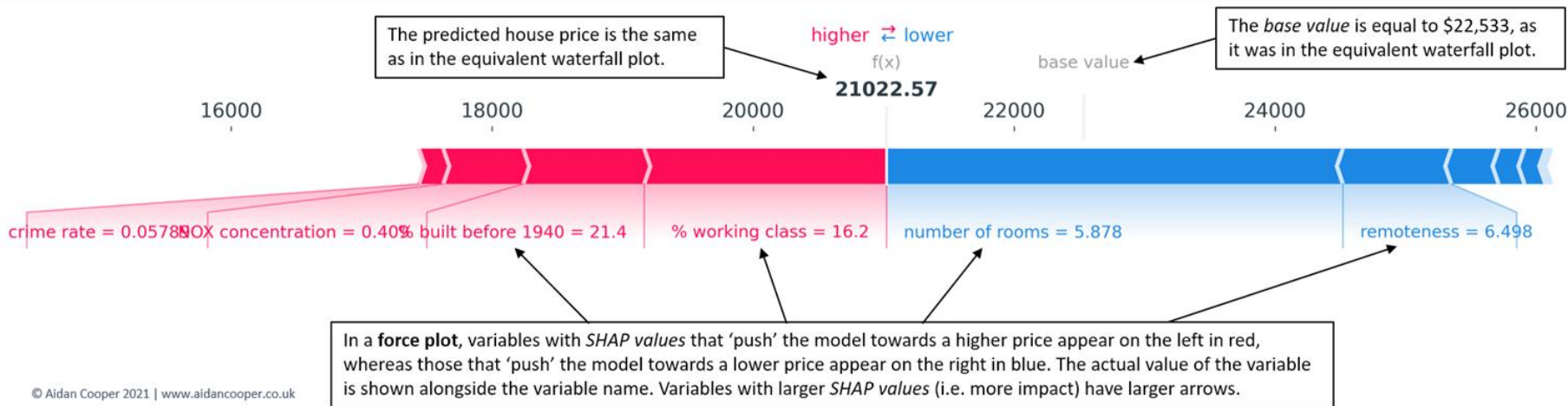


The SHAP values quantify the amount and direction in which each variable impacts the predicted house price.

SHAP values shown inside red arrows correspond to input variables that 'push' the model towards predicting a higher price, whereas those in blue 'push' the model towards a lower price.

The final prediction, $f(x)$, is equal to the base value plus the sum of all the SHAP values.

The base value is the same for all examples in the data. It is equal to the average house price in the dataset.



80th Percentile House Price



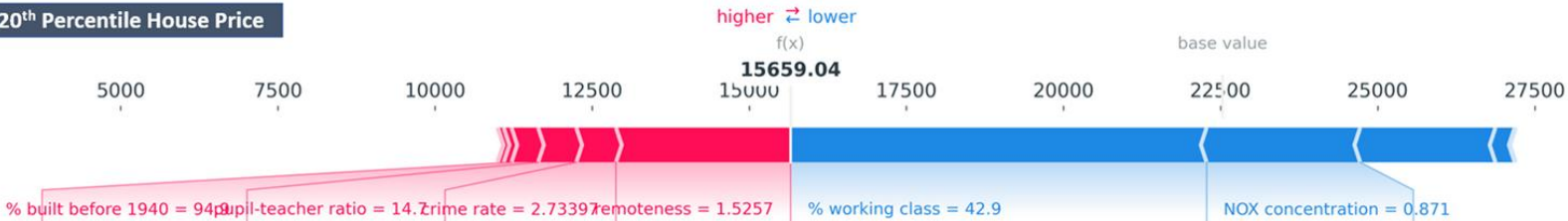
60th Percentile House Price



40th Percentile House Price

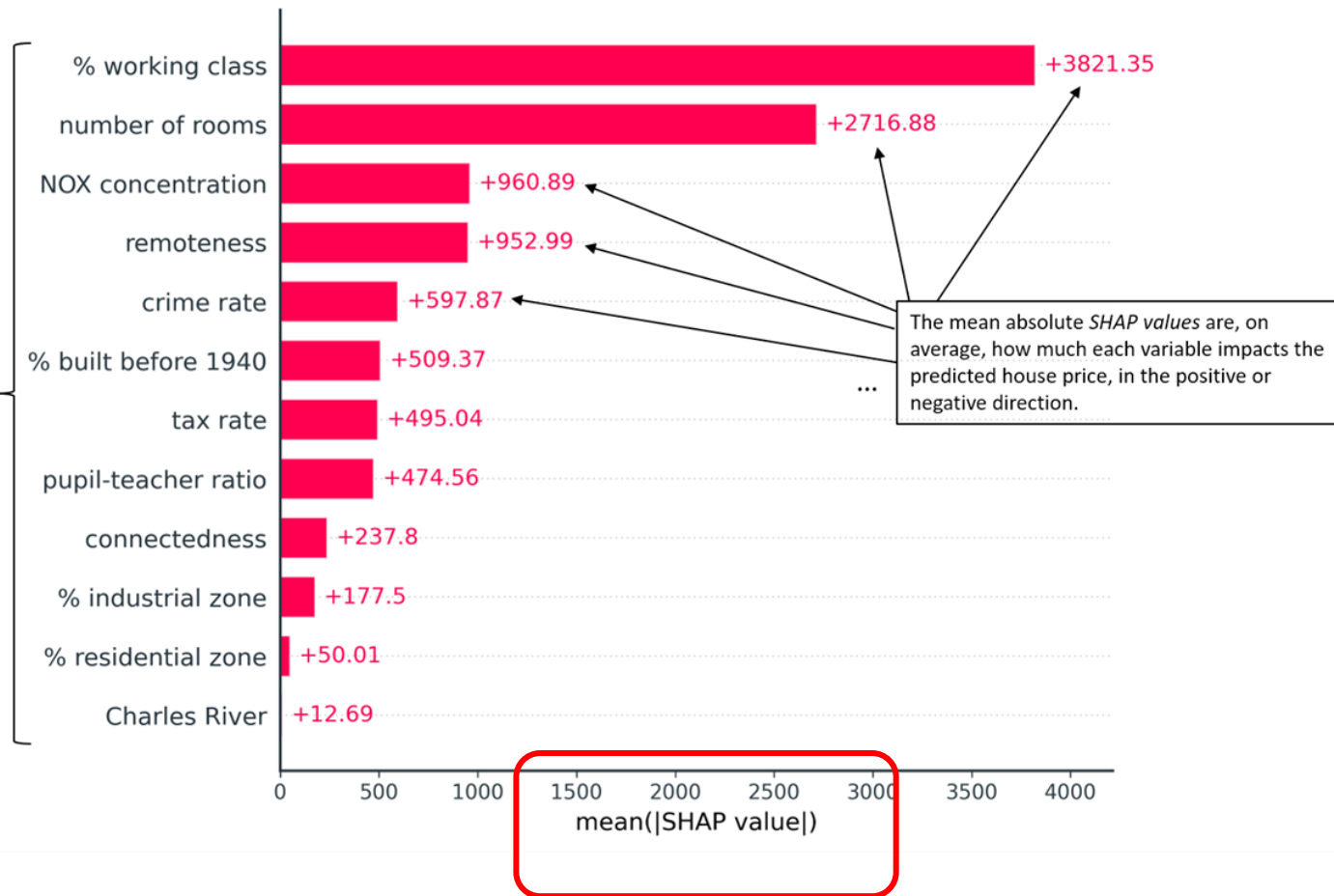


20th Percentile House Price



**Interpretabilidad global: sobre
todo el conjunto de test**

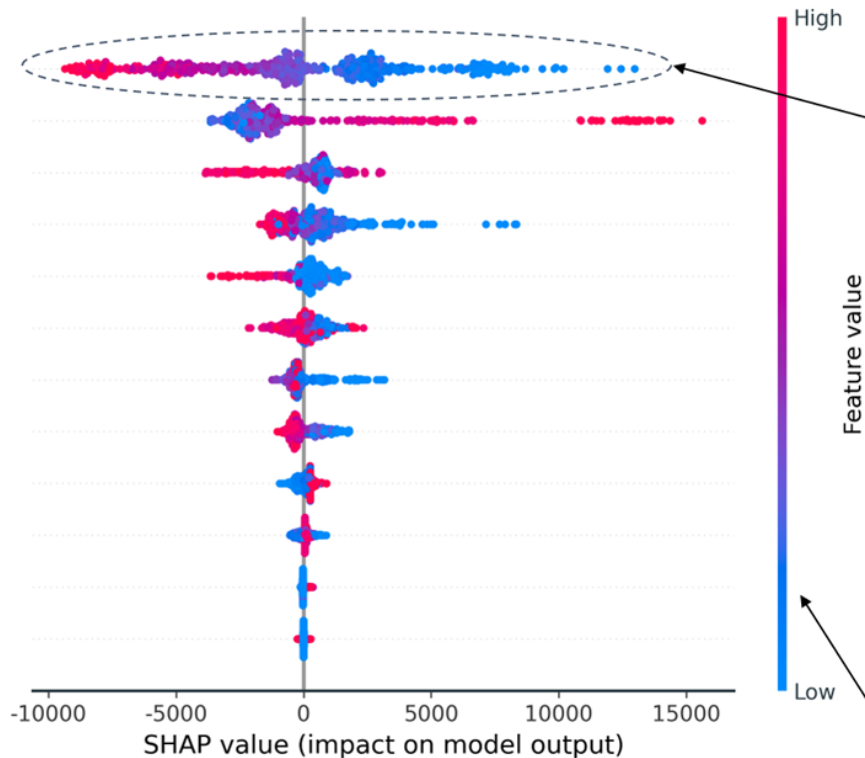
These are the input variables, ranked from top to bottom by their mean absolute *SHAP values* for the entire dataset – i.e. the average magnitude of each variable's impact on the predicted house price across all instances.



% working class
 number of rooms
 NOX concentration
 remoteness
 crime rate
 % built before 1940
 tax rate
 pupil-teacher ratio
 connectedness
 % industrial zone
 % residential zone
 Charles River

These are the input variables, ranked from top to bottom by their mean absolute *SHAP* values for the entire dataset.

Note: this ranking is exactly the same as for the bar plot.



In a **beeswarm plot**, for each variable, every instance (i.e. row) of the dataset appears as its own point. The points are distributed horizontally along the x-axis according to their *SHAP* value. In places where there is a high density of *SHAP* values, the points are stacked vertically.

Examining how the *SHAP* values are distributed reveals how a variable may influence the model's predictions.

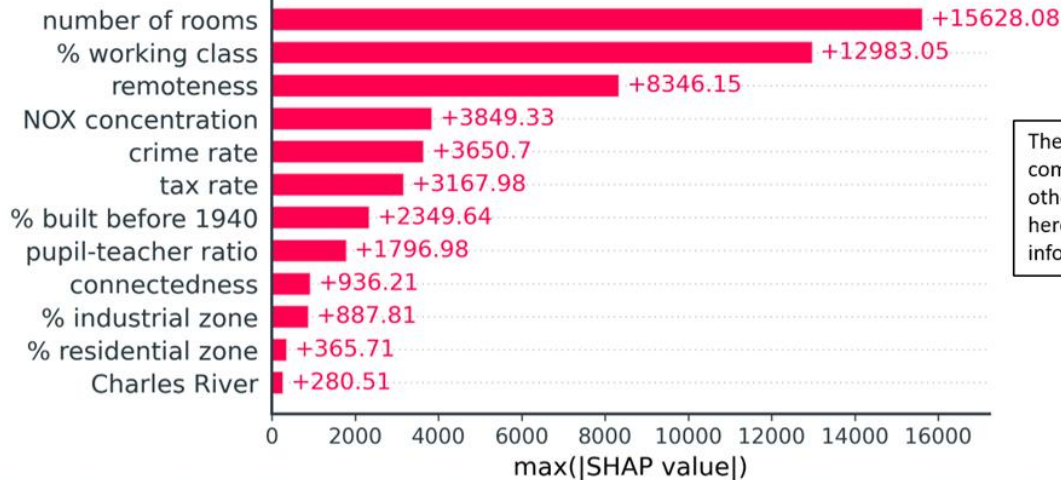
The colour bar corresponds to the raw values (not to be confused with the *SHAP* values) of the variables for each instance (i.e. point) on the graph.

If the value of a variable for a particular instance is relatively high, it appears as a red dot. Relatively low variable values appear as blue dots.

Examining the colour distribution horizontally along the x-axis for each variable provides insights into the general relationship between a variable's raw values and its *SHAP* values.

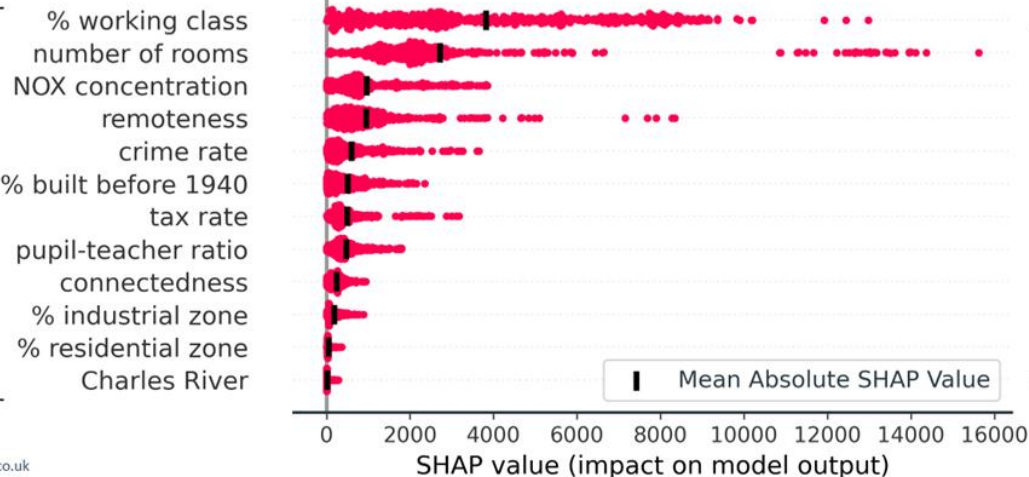
Whereas before, variables were ranked by their **mean** absolute *SHAP* value, here they are ranked by their **max** absolute *SHAP* value for the entire dataset.

Note that the ranking changes in places.



The mean absolute *SHAP* value is the most commonly used ranking for variables, but other statistics such as the max value (shown here) or median value may also be informative.

These are the input variables, ranked from top to bottom by their mean absolute *SHAP* values for the entire dataset.



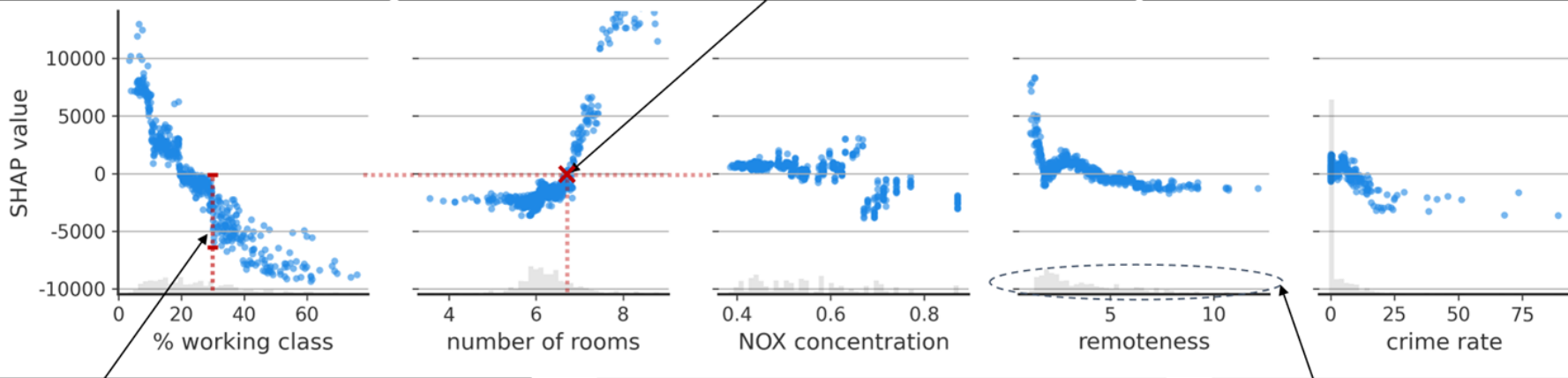
A balance can be struck between the simplicity of a bar plot and the information-rich complexity of a beeswarm plot, by creating a beeswarm plot for the **absolute** *SHAP* values.

This still shows us the ranking and relative influence of variables on the model's predictions, but also allows for further insights. E.g. the highest observed *SHAP* values actually occur for the 2nd ranked variable, *number of rooms*.

In a **dependence plot**, every instance (i.e. row) of the dataset appears as it's own point. The points are presented as a scatterplot of a variable's *SHAP values* versus the variables underlying raw values.

SHAP values above the $y=0$ line lead to predictions of higher house prices, whereas those below it are associated with lower house price predictions. The raw variable value at which the distribution of *SHAP values* cross the $y=0$ line tells you the threshold at which the model switches from predicting lower to higher house prices. For *number of rooms*, this is at approximately 6.8 rooms, as marked by the **X**.

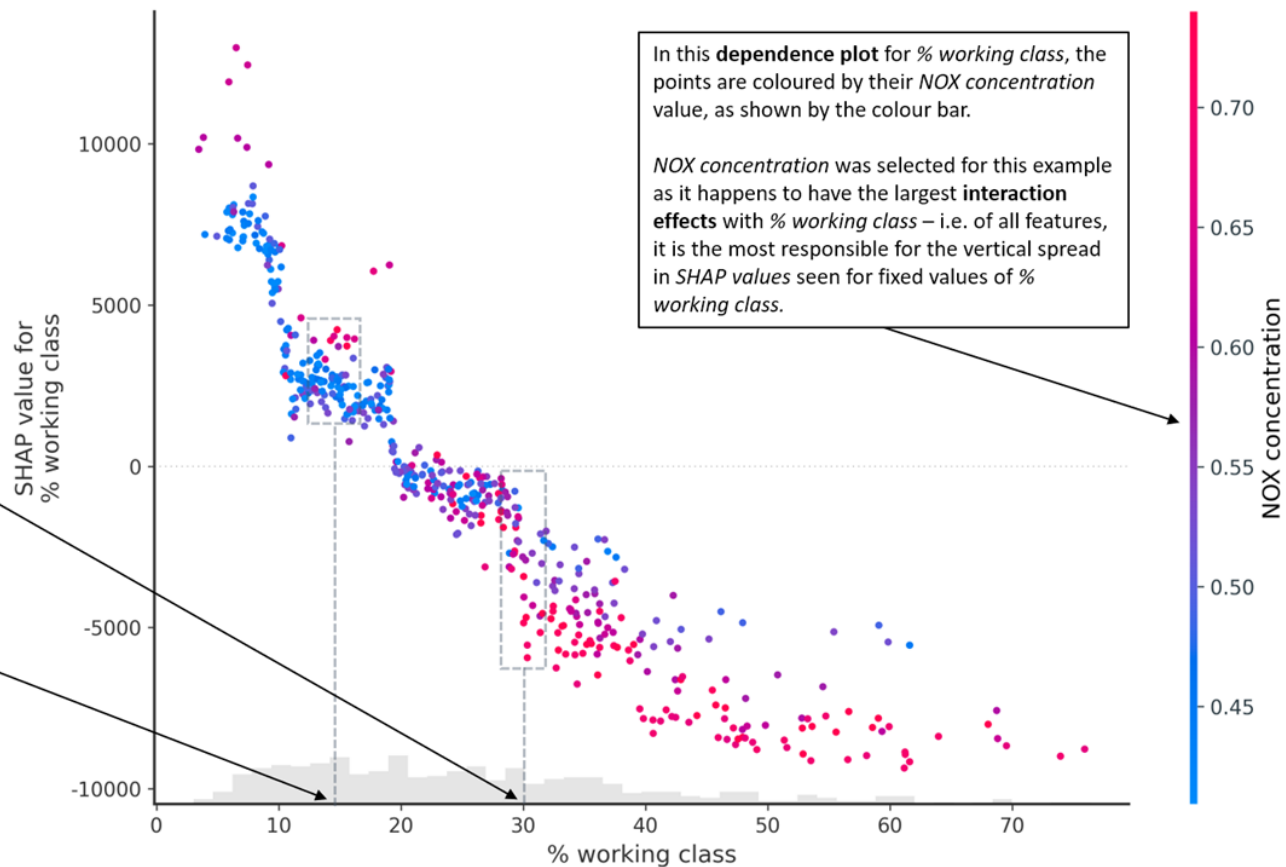
With all five plots on the same y-scale, the extent of the vertical distribution of the *SHAP values* indicates how much relative influence each variable has on predictions. *% working class* has a much wider range of *SHAP values* than *crime rate*.



The vertical spread of *SHAP values* at a fixed raw variable value is due to *interaction effects* with other variables. For example, here we see that houses with a *% working class* of 30% can have *SHAP values* that range from \$0 to -\$6,500 depending on the other data for those particular instances.

The shapes of the distributions of points provide insights into the relationship between a variable's values and its *SHAP values*. For *% working class*, we see a negative, linear relationship across the full range of variable values. For *number of rooms*, we see that *SHAP values* are mostly flat between 4 and 6.5 rooms, but then increase sharply for higher room counts.

The inset histograms just above the x-axis display the distributions of raw variable values. We should be cautious not to overinterpret regions of the dependence plot where the underlying data is sparse (e.g. *crime rates* over 25%).



The *SHAP values* at a % working class value of 30% range from roughly \$0 to -\$6,500. Examining the colour distribution, it can be seen that higher levels of *NOX concentration* are associated with the decrease in *SHAP values* for this region.

At a % working class value of 15%, the *SHAP values* range from roughly \$1,000 to \$5,000. For this region, it can be seen that instances with the highest *NOX concentration* values have the highest *SHAP values*, whereas instances with lower *NOX concentration* values have lower *SHAP values*. This is the inverse relationship to that seen at 30% working class.



Bibliografía

<https://shap.readthedocs.io/en/latest/index.html>

<https://www.aidancooper.co.uk/a-non-technical-guide-to-interpreting-shap-analyses>